

Small signal stability analysis and control of the wind turbine with the direct-drive permanent magnet generator integrated to the grid

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ABSTRACT

In this paper, a model of the wind turbine (WT) with direct-drive permanent magnet generator (DDPMG) and its associated controllers is presented, based on which a small signal stability analysis model is derived. The small signal stability analysis shows that the WT with DDPMG without the controllers is stable, and the controller can improve the small signal stability of the WT system. To evaluate the control capability of the proposed controller, simulations are performed on a single machine infinite bus system and a four-machine system, and the dynamic responses of the DDPMG are compared with those of the WT with doubly fed induction generator (DFIG). Under the system disturbance, the dynamic performances of the DDPMG are better than those of the WT with DFIG. The impact of the WT with DDPMG and the WT with DFIG on the transient stability of the power system is also investigated in term of critical clear time (CCT). The transient stability margin of the system can be improved with the integration of the WT systems, while between these two WT systems, the stability of the power system with the WT with DDPMG is further enhanced.

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1. Introduction

The wind power generation has been developed very quickly in the past few years [1]. With the growing penetration of wind energy generation into power grids, the impact of wind turbine (WT) on power system operations and stability control is of increasing concern.

WT system normally consists of wind turbine, generator and grid interface converters, among which the generator is an very important component in the WT system. During the development of the WT techniques, synchronous generator, induction generator and doubly fed induction generator were employed to convert wind power to electrical power. Wind turbines usually rotate at a speed of 15–20 rev/min, and generators should rotate at the speed of 1000–1500 rev/min. Hence, a gearbox should be connected between a wind turbine and a generator. The gearbox causes unpleasant noise, increases the loss of the WT system, and also requires regular maintenance. In order to overcome these problems, the WT with direct-drive permanent magnet generator (DDPMG), where the gearbox was eliminated, was developed [2]. The permanent magnet generator driven directly by the wind turbine is a multi-pole and low speed generator. Different types of DDPMG have been developed

for the WT system, such as axial-flux machine and radial-flux machine [3–6].

The WT with DDPMG was compared with the other types of WT systems in [7], such as the WT with doubly-fed induction generator (DFIG) with three-stage gearbox, the WT with DFIG with single gearbox, the WT with DDPMG with single gearbox, and the WT with direct-drive synchronous generator with electrical excitation. The results have shown that the WT with DDPMG has the highest energy yield, and is considered as the best solution for wind energy conversion systems since it does not need brushes or a gearbox, while it can be interfaced with the grid through full-scale back-to-back converters. However, the comparisons made in [7] are only concerned with the characteristics of the WT system themselves, while the impact of different WT configurations on the dynamic responses or the stability of the power system has not been addressed.

Since the rotor flux linkage of DDPMG is established by the permanent magnet and the rotor winding is eliminated, the control strategy for the WT with DDPMG is different to that of the WT equipped with conventional generators. A control strategy for the WT with DDPMG was proposed in [8], the generator side converter controller is designed to maintain the rotating speed of the DDPMG at the optimal rotating speed, and minimize the core power losses; the grid side converter controller is designed to maintain the voltage of the DC-link, and control the output reactive power to the specified reference. The similar control strategy was also presented in [9], the output active power and reactive power of the DDPMG was controlled by the generator side converter controller, while

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the DC-link voltage and the terminal voltage of the WT system was controlled by the grid side converter controller. A maximum power tracking control strategy without mechanical sensor was proposed in [10], and compared with the maximum power tracking control strategy previously proposed, the maximum power tracking capability can be improved using the proposed control strategy. The control strategies mentioned above were designed to improve the dynamic responses of the WT system itself or increase the output energy of the wind power system. However, their control capabilities have not been evaluated under the system disturbances and their impact on power grid operations and stability has not been addressed.

In this paper, a simple control strategy for the WT with DDPMG is proposed. The detailed model of WT with DDPMG including its associated converters and controllers is presented. Based on which the small signal stability analysis model is derived, and the small signal stability analysis is then carried out. Dynamic simulations are performed on a single machine infinite bus (SMIB) system and a four-machine system to simulate the dynamics of the WT with DDPMG and to evaluate the control capability of the proposed controller under the system disturbances. The dynamic responses of the WT with DDPMG are compared with those of the WT with DFIG, and the critical clear times (CCTs) of the four-machine system with the WT with DDPMG and the four-machine system with the WT with DFIG are also compared to illustrate the impact of the different WT configurations on the stability of the power system.

2. Model of wind turbine with DDPMG

2.1. Model of DDPMG

Basically, DDPMG is a synchronous generator with constant rotor flux linkage, the equivalent circuit is shown in Fig. 1. Aligning the direction of the d axis of the $d-q$ reference frame with the flux linkage, the model of the DDPMG is given by

$$\begin{aligned} L_s \frac{di_{ds}}{dt} &= -v_{ds} - R_s i_{ds} + L_s \omega i_{qs} \\ L_s \frac{di_{qs}}{dt} &= -v_{qs} - R_s i_{qs} - L_s \omega i_{ds} + \omega \psi \end{aligned} \quad (1)$$

where v_{ds} and v_{qs} are the direct (d) and quadrature (q) axis stator voltage, respectively; i_{ds} and i_{qs} are the d and q axis stator current, respectively; R_s is the resistance of the stator; L_s is the inductance of the stator; ω is the generator electrical speed; ψ is the magnet flux linkage.

Power equations are given by:

$$P_s = v_{ds} i_{ds} + v_{qs} i_{qs} \quad (2)$$

$$Q_s = v_{qs} i_{ds} - v_{ds} i_{qs} \quad (3)$$

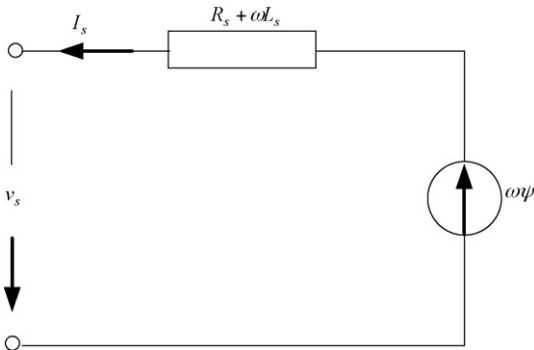


Fig. 1. Equivalent circuit of DDPMG.

where P_s and Q_s are the output active power and reactive power of the DDPMG, respectively.

2.2. Model of drive train

Since the WT with DDPMG is normally decoupled with the power grid by the full scale back-to-back converters, the transients in the power grid will not be seen at the DDPMG terminal and the shaft system. Hence, for the model of the WT with DDPMG integrated into the power grid, the mechanical system of the WT can be simply modeled with the one-mass [11,12] model given by

$$H_{tot} \frac{d\omega_t}{dt} = T_M - T_e \quad (4)$$

$$H_{tot} = H_t + H_g \quad (5)$$

$$\omega_t = \frac{\omega}{p} \quad (6)$$

where ω_t is the turbine speed; H_{tot} is the inertia constant of the whole drive train; H_t and H_g is the inertia constant of the turbine and the generator, respectively; p is the pole pair number; T_M is the mechanical torque of the turbine; T_e is the generator electromagnetic torque. T_m and T_e are given by

$$T_m = \frac{0.5 \rho \pi R^2 C_p V_w^3}{\omega_t} \quad (7)$$

$$T_e = -p \psi i_{qs} \quad (8)$$

where

$$C_p = \frac{1}{2} \left(\frac{R C_f}{\lambda} - 0.022 \beta - 2 \right) e^{-0.255 (R C_f / \lambda)} \quad (9)$$

where ρ is the air density; R is the wind turbine blade radius; V_w is the wind speed; C_f is the blade design constant coefficient; β is the blade pitch angle; λ is the blade tip speed ratio, $\lambda = \omega_t R / V_w$; C_p is the power coefficient.

2.3. Model of converters

The WT with DDPMG is interfaced with the power grid via full scale back-to-back converters. The configuration of the converters is shown in Fig. 2. Since the active power flow through the converters is balanced via the DC-link, with the definition of the positive direction of the currents in Fig. 2, the power balance equation can be obtained as follows:

$$0 = P_{DC} + P_g + P_s \quad (10)$$

where P_{DC} is the active power of the DC-link; P_g is the active power at the AC terminal of the grid side converter. They are given by

$$P_{DC} = v_{DC} i_{DC} = -C v_{DC} \frac{dv_{DC}}{dt} \quad (11)$$

$$P_g = v_{Dg} i_{Dg} + v_{Qg} i_{Qg} \quad (12)$$

where i_{Dg} and i_{Qg} are the D and Q axis currents of the grid side converter, respectively; v_{Dg} and v_{Qg} are the D and Q axis voltages of the grid side converter, respectively; v_{DC} is the terminal voltage of

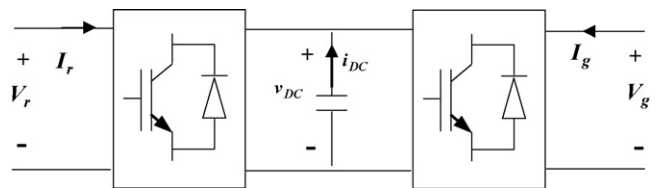


Fig. 2. Full scale back to back converters.

capacitor; i_{DC} is the current of the capacitor; C is the capacitance of the capacitor.

Since the AC systems of the generator and the grid are decoupled by the DC-link, they can operate at different AC frequencies. Hence, $d-q$ and $D-Q$ references of frame are used in the modeling of the generator and the grid, respectively.

The model of the converters and DC-link can be derived as follows:

$$C v_{DC} \frac{dv_{DC}}{dt} = v_{Dg} i_{Dg} + v_{Qg} i_{Qg} + v_{ds} i_{ds} + v_{qs} i_{qs} \quad (13)$$

3. Model of controllers

3.1. Generator side converter controller model

The generator side converter controller aims to control the output active power of the generator to track the input of the wind turbine torque, and minimize the power losses in the generator. The active power is controlled via v_{qs} . The power losses of the generator are minimized by controlling i_{ds} to zero. The control is implemented via v_{ds} . The control block diagrams are shown in Fig. 3, while control equations are given by

$$\frac{dx_1}{dt} = P_s - P_{ref} \quad (14)$$

$$\frac{dx_2}{dt} = i_{ds} - i_{ds_ref} \quad (15)$$

$$v_{qs} = K_{p1} \Delta P + K_{i1} x_1 - L_s \omega i_{ds} + \omega \psi \quad (16)$$

$$v_{ds} = K_{p2} \Delta i_{ds} + K_{i2} x_2 + L_s \omega i_{qs} \quad (17)$$

where x_1, x_2 are the intermediate variables; K_{p1} and K_{i1} are the proportional and integrating gains of the power regulator, respectively; K_{p2} and K_{i2} are the proportional and integrating gains of the d axis stator current regulator, respectively; i_{ds_ref} is the specified d axis stator current reference; P_{ref} is the active power control reference.

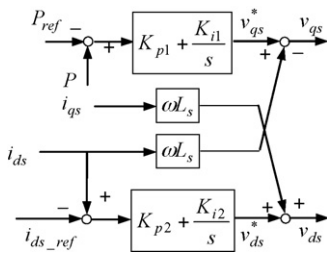


Fig. 3. Generator side converter control block diagram.

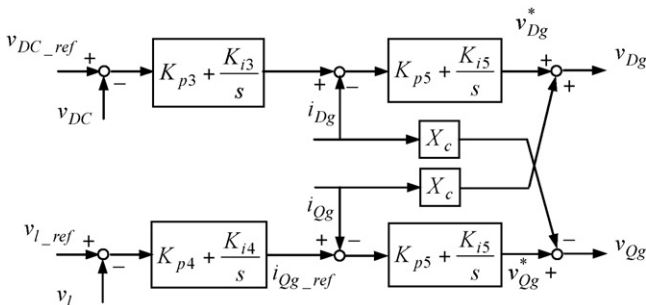


Fig. 4. Grid side converter control block diagram.

3.2. Grid side converter controller

The grid side converter controller, as shown in Fig. 4, aims to maintain the DC-link voltage and the terminal voltage of the WT system. Aligning the direct (D) axis of the $D-Q$ reference frame with the terminal voltage of the WT system, the voltage of the DC-link and the terminal voltage of the WT system are controlled independently via v_{Dg} and v_{Qg} , respectively [13]. Introducing the intermediate variables x_3, x_4, x_5 and x_6 , the following control equations of the controller shown in Fig. 4 can be obtained:

$$\frac{dx_3}{dt} = v_{DC_ref} - v_{DC} \quad (18)$$

$$\frac{dx_4}{dt} = i_{Dg_ref} - i_{Dg} = -K_{p3} \Delta v_{DC} + K_{i3} x_3 - i_{Dg} \quad (19)$$

$$\frac{dx_5}{dt} = V_{t_ref} - V_t \quad (20)$$

$$\frac{dx_6}{dt} = i_{Qg_ref} - i_{Qg} = -K_{p4} \Delta v_t + K_{i4} x_5 - i_{Qg} \quad (21)$$

$$\begin{aligned} v_{Dg} &= K_{p5} \frac{dx_4}{dt} + K_{i5} x_4 \\ &= K_{p5} (-K_{p3} \Delta v_{DC} + K_{i3} x_3 - i_{Dg}) + K_{i5} x_5 + X_c i_{Qg} \end{aligned} \quad (22)$$

$$\begin{aligned} v_{Qg} &= K_{p5} \frac{dx_6}{dt} + K_{i5} x_6 \\ &= K_{p5} (-K_{p4} \Delta v_t + K_{i4} x_5 - i_{Qg}) + K_{i5} x_6 - X_c i_{Dg} \end{aligned} \quad (23)$$

where K_{p3} and K_{i3} are the proportional and integrating gains of the DC bus voltage regulator, respectively; K_{p4} and K_{i4} are the proportional and integrating gains of the terminal voltage regulator, respectively; K_{p5} and K_{i5} are the proportional and integrating gains of the grid-side converter current regulator, respectively; v_{DC_ref} is the voltage control reference of the DC-link; V_t is the magnitude of the terminal voltage of the WT system, $V_t = \sqrt{v_{Dl}^2 + v_{Ql}^2}$; v_{Dl} and v_{Ql} are the D and Q axis components of the terminal voltage of the WT system, respectively; V_{t_ref} is the control reference for the terminal voltage of the WT system; X_c is the grid side converter smoothing reactance.

4. Dynamic model of WT with DDPMG system

Eqs. (1), (4) and (13) show the model of the WT with DDPMG system, among which (1) and (4) describe the electrical dynamics of the WT with DDPMG system, while (13) describes the mechanical dynamics of the WT with DDPMG system. The models of the controllers are given by Eqs. (14)–(23). The dynamic model of the WT with DDPMG may be written in the compact form as follows:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{z}, \mathbf{u}) \quad (24)$$

$$\mathbf{z} = \mathbf{g}(\mathbf{x}, \mathbf{u}) \quad (25)$$

where $\mathbf{x}, \mathbf{z}, \mathbf{u}$ are the vectors of the DDPMG state variables, control output variables and input variables, respectively, and they are given by

$$\begin{aligned} \mathbf{x} &= [\omega_t, i_{ds}, i_{qs}, v_{DC}, x_1, x_2, x_3, x_4, x_5, x_6]^T, \quad \mathbf{z} = [v_{ds}, v_{qs}, v_{Dg}, v_{Qg}]^T, \\ \mathbf{u} &= [v_{Dl}, v_{Ql}, i_{Dg}, i_{Qg}]^T \end{aligned}$$

5. Small signal model of WT with DDPMG

5.1. Interfacing WT with DDPMG with power grid

Taking the single machine and infinite bus system shown in Fig. 5a as an example, the network voltage equation can be obtained

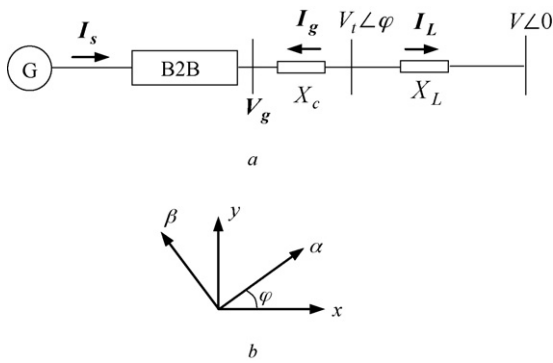


Fig. 5. Interface with power grid.

as follows:

$$V_t \angle \varphi - V \angle 0 = jX_L I_L = -jX_L I_g \quad (26)$$

where V is the voltage of the infinite bus; $V_t \angle \varphi$ is the terminal voltage of the WT system; X_L is the reactance of the transmission line.

The network voltage equation (26) and the grid side converter model of the WT system are presented in x - y and D - Q frame of references, respectively. The relationship between these two references frame is illustrated in Fig. 5b, while the corresponding transformation between these two coordinates is given by (27)

$$\begin{bmatrix} f_D \\ f_Q \end{bmatrix} = \begin{bmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{bmatrix} \begin{bmatrix} f_x \\ f_y \end{bmatrix} \quad (27)$$

where f_D and f_Q are the D and Q axis components, respectively; f_x and f_y are the x and y axis components, respectively.

Applying (27) to (26), the network voltage equation can be written as

$$\begin{bmatrix} v_{Dl} \\ v_{Ql} \end{bmatrix} = \begin{bmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{bmatrix} V - \begin{bmatrix} 0 & -X_L \\ X_L & 0 \end{bmatrix} \begin{bmatrix} i_{Dg} \\ i_{Qg} \end{bmatrix} \quad (28)$$

The voltage equation of the grid side converter is

$$\begin{bmatrix} v_{Dl} \\ v_{Ql} \end{bmatrix} - \begin{bmatrix} v_{Dg} \\ v_{Qg} \end{bmatrix} = \begin{bmatrix} 0 & -X_c \\ X_c & 0 \end{bmatrix} \begin{bmatrix} i_{Dg} \\ i_{Qg} \end{bmatrix} \quad (29)$$

Linearizing (28) and (29), we have:

$$\begin{bmatrix} \Delta v_{Dl} \\ \Delta v_{Ql} \end{bmatrix} = \begin{bmatrix} 0 & X_L \\ -X_L & 0 \end{bmatrix} \begin{bmatrix} \Delta i_{Dg} \\ \Delta i_{Qg} \end{bmatrix} \quad (30)$$

$$\begin{bmatrix} \Delta v_{Dl} \\ \Delta v_{Ql} \end{bmatrix} - \begin{bmatrix} \Delta v_{Dg} \\ \Delta v_{Qg} \end{bmatrix} = \begin{bmatrix} 0 & -X_c \\ X_c & 0 \end{bmatrix} \begin{bmatrix} \Delta i_{Dg} \\ \Delta i_{Qg} \end{bmatrix} \quad (31)$$

Combining (30) and (31), the following equation can be obtained:

$$\Delta \mathbf{u} = K_1 \Delta \mathbf{x} + K_2 \Delta \mathbf{z} \quad (32)$$

5.2. Small signal stability model

Linearizing (24) and (25), we have

$$\Delta \dot{\mathbf{x}} = A \Delta \mathbf{x} + B \Delta \mathbf{z} + C \Delta \mathbf{u} \quad (33)$$

$$\Delta \dot{\mathbf{z}} = D \Delta \mathbf{x} + E \Delta \mathbf{u} \quad (34)$$

Substituting (32) and (34) into (33), small signal model of the WT with DDPMG can be obtained:

$$\Delta \dot{\mathbf{x}} = \mathbf{A} \Delta \mathbf{x} \quad (35)$$

Table 1

Eigenvalues and participation factors of WT without controllers.

λ	σ	ω	F (Hz)	ASV ₁	PF ₁	ASV ₂	PF ₂
$\lambda_{1,2}$	-183.59	314.43	50.1	i_q	0.5002	i_d	0.4994
λ_3	-0.87	0	0	v_{DC}	1.0	-	-
λ_4	-0.32	0	0	ω	0.9991	-	-

ASV: associated state variable.

Table 2

Eigenvalues and participation factors of WT with controllers.

λ	σ	ω	f (Hz)	ASV ₁	PF ₁	ASV ₂	PF ₂
$\lambda_{1,2}$	-583.63	109.07	17.36	v_{DC}	0.4666	x_6	0.2974
$\lambda_{3,4}$	-99.04	89.64	14.27	i_q	0.5	x_1	0.4999
λ_5	-0.23	0	0	ω	0.9998	-	-
λ_6	-91.10	0	0	x_4	0.9084	-	-
λ_7	-47.56	0	0	x_5	0.6751	x_6	0.3519
λ_8	-25.02	0	0	x_3	0.9982	-	-
λ_9	-523.1	0	0	i_d	1.0568	-	-
λ_{10}	-28.10	0	0	x_2	1.0568	-	-

where $\mathbf{A}$ is the system state matrix, whose dimension is 10×10

$$\Delta \mathbf{x} = [\Delta \omega_t, \Delta i_{ds}, \Delta i_{qs}, \Delta v_{DC}, \Delta x_1, \Delta x_2, \Delta x_3, \Delta x_4, \Delta x_5, \Delta x_6].$$

6. Small signal stability analysis

6.1. WT with DDPMG without controller

The linearized model (35) is used for the small signal stability analysis. The eigenvalues and participation factors of the system state matrix can be calculated using the parameters presented in Appendix A. The eigenvalue analysis results of the WT with DDPMG without controller are shown in Table 1, and it can be seen that the WT with DDPMG without controller is stable since all the eigenvalues have negative real parts. In particular there is one mode of oscillation with a frequency of 50.1 Hz. The oscillation mode is associated with the stator current of the generator.

6.2. WT with DDPMG considering controllers

Small signal stability analysis for the WT with DDPMG considering controllers was carried out using the controller parameters listed in Appendix B. The analysis results are shown in Table 2. It can be found that when the controller designed is applied, the system is stable. In comparison to the results in Table 1 where WT controllers are not considered, the system damping increases and there are two modes of oscillation: one is mainly associated with V_{DC} with a frequency of 17.36 Hz; the other one is associated with i_q and x_1 with a frequency of 14.27 Hz. The frequencies of the oscillation modes of the WT with controller are much smaller than that of the WT without controller.

7. Dynamic simulations

The simulations were performed to evaluate the control capability of the proposed controller under the system fault in MATLAB/SIMULINK, and the dynamic responses of the WT with DDPMG were compared with those of the WT with DFIG to illustrate the impact of the wind generation with different configurations on the stability of the power systems, the details of the WT with DFIG can be found in [14]. A single machine infinite bus (SMIB) system and a 4-machine system were used in the dynamic simulations.

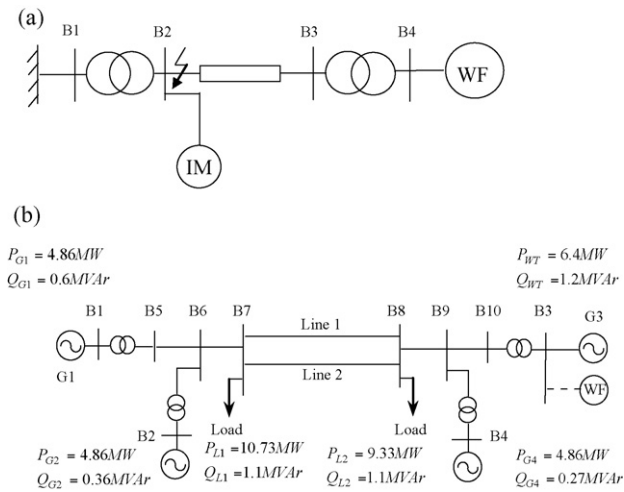


Fig. 6. Configuration of the simulation system.

7.1. Simulations on SMIB system

The SMIB system consists of a wind farm including six wind turbines with a total capacity of 9 MW, a motor load, and an infinite voltage source, the wind farm is connected with the infinite voltage bus through a 10 km transmission line, the load is located at the terminal bus of the transmission line near the infinite voltage source, the configuration of the SMIB system is shown in Fig. 6a. A three phases ground fault with the fault resistance of 11Ω was applied at the transmission line terminal bus near the infinite voltage source at $t = 0+$ and cleared after 0.05 s. The dynamic responses of the WT with DDPMG and the WT with DFIG under the disturbance are shown in Fig. 7. It can be seen that with the proposed

control, the WT with DDPMG system is stable and the dynamic responses of the WT system with DDPMG are better than those of the WT with DFIG. The oscillations of the terminal voltage of the WT with DDPMG were damped quickly, and the peak DC-link voltage of the WT with DDPMG is a little bit smaller than that of the WT with DFIG. During the transient process, the minimum active power of the WT with DDPMG is 1.9 MW where the WT generated the active power, while the minimum active power of the WT with DFIG is -3.5 MW where the WT absorbed the active power. Under the disturbance, the WT with DDPMG generated more reactive power to support the terminal voltage since it is interfaced with the power grid via full scale back-to-back converters. However, the full scale converters are more expensive than the fed back converters used in WT with DFIG system.

7.2. Simulations on four-machine system

The four-machine system used in the simulations is shown in Fig. 6b. The system was divided into two areas, each of which consists of two machines. To investigate the impact of the WT with different configurations (WT with DDPMG and WT with DFIG) on the transient stability margin of the power system, the four-machine system was modified by replacing generator G3 with a wind farm (WF), which is the same as the WF in Fig. 6a. The power flow of the four-machine system is also shown in Fig. 6b.

7.2.1. Case 1

A three phase ground fault with fault resistance of 0.001Ω appeared at bus 8 at $t = 0+$. The CCT was calculated under different conditions such as the original four-machine system, the modified four-machine system with the WT with DFIG and the modified four-machine system with the WT with DDPMG. The results are shown in Table 3.

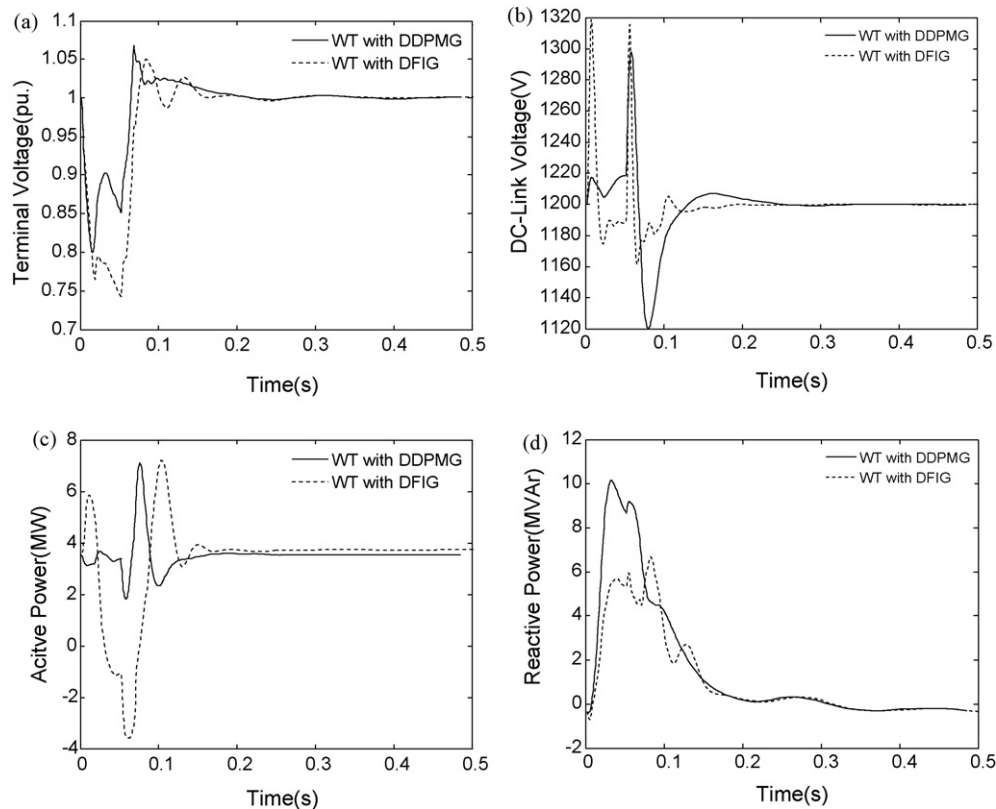


Fig. 7. Dynamics of the WT with DDPMG.

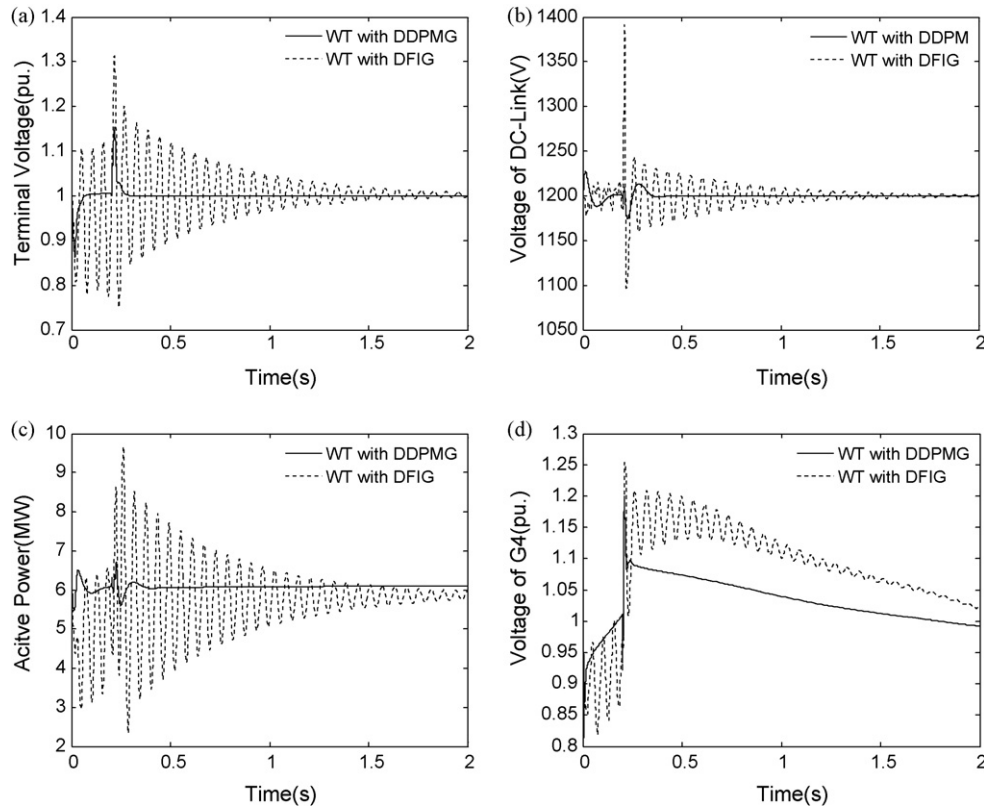


Fig. 8. Dynamic responses of the four-machine system.

Table 3

The CCT of the simulated system under different conditions.

Controllers	Original 4-machine system	Modified system	
		WT with DFIG	WT with DDPMG
CCT(s)	0.55	0.567	0.6

From Table 3, it can be seen that the transient stability of multi-machine system can be enhanced by the integration both of the WT with DDPMG and WT with DFIG. Furthermore, for the four-machine systems with the WT system, the CCT of the system with the WT with DDPMG is higher than that of the system with the WT with DFIG.

7.2.2. Case 2

A three phase ground fault with fault resistance of 0.001 Ω appeared at the middle of the transmission line 1 at $t=0^+$, and was cleared after 0.2 s, simulations were carried out, and dynamics of the terminal voltage, the DC-link voltage, and the output active power of the WT with DDPMG and the WT with DFIG are shown in Fig. 8a–c, respectively. It can be seen that the dynamic responses of the WT with DDPMG are better than those of the WT with DFIG and the oscillations were damped very quickly. The dynamic responses of G4, which is located at the same area of the WF, were significantly affected. The dynamics of the terminal voltage of G4 is also shown in Fig. 8d. It can also be seen that the oscillations of the terminal voltage of G4 were damped quickly for the power system with the WT with DDPMG.

8. Conclusion

In the paper, a dynamic model of the WT with DDPMG including its converters and controllers has been presented, based on which the model for small signal stability analysis has been estab-

lished. The small signal stability analysis of the WT system with and without controllers has also been carried out. It has been found that the WT with DDPMG system without the controllers is stable, while using the controllers the dynamic performance of the WT system can be improved. The dynamic responses of the WT with DDPMG were simulated on the single machine system and the four-machine system, and comparisons with those of WT with DFIG were also made. The simulation results have illustrated that in comparison to the WT with DFIG, the WT with DDPMG has better dynamic responses. The CCT analysis has further showed that both the WTs can improve the stability margin of the power system. Between these two WTs, the WT with DDPMG has shown even better transient stability performance.

Appendix A. Parameters of WT with DDPMG

Wind turbine:

$$H_t = 5 \text{ s}, H_g = 0.5 \text{ s}, R = 50 \text{ m}, C_f = 0.733$$

PMG:

$$R_s = 0.01 \text{ pu}, L_s = 0.171 \text{ pu}, \psi = 1.1 \text{ pu}$$

Converter:

$$C = 0.001 \text{ F}, V_{DC0} = 1200 \text{ V}, X_c = 0.55 \text{ pu}$$

Wind:

$$V_w = 12 \text{ m/s}$$

Appendix B. Controller parameters

$$k_{p1} = 0.3 \text{ pu}, k_{i1} = 100 \text{ s}^{-1}, k_{p2} = 0.5 \text{ pu}, k_{i2} = 8 \text{ s}^{-1}, k_{p3} = 0.002 \text{ pu}, k_{i3} = 0.05 \text{ s}^{-1}, k_{p4} = 1.25 \text{ pu}, k_{i4} = 300 \text{ s}^{-1}, k_{p5} = 0.1 \text{ pu}, k_{i5} = 10 \text{ s}^{-1}.$$

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